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DYNAMIC SEARCH FOR A MOVING TARGET

DAVID ASSAF AND
ARIELA SHARLIN-BILITZKY,* *The Hebrew University of Jerusalem*

Abstract

An object is hidden in one of two boxes and occasionally moves between the boxes in accordance with some specified continuous-time Markov process. The objective is to find the object with a minimal expected cost. In this paper it is assumed that search efforts are unlimited. In addition to the search costs, the 'real time' until the object is found is also taken into account in the cost structure. Our main results are that the optimal policy may consist of five regions and that the controls applied should be of the extreme 0 or ∞ type. The resulting expected cost compares favorably with that of the expected cost with bounded controls studied previously in the search literature.

OPTIMAL SEARCH POLICY; DYNAMIC PROGRAMMING; OPTIMALITY EQUATION; MARKOV CHAINS WITH CONTINUOUS PARAMETER

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1. Introduction and summary

The basic continuous-time search problem may be described as follows. An 'object' (or 'target') is hidden in one of m 'boxes' labeled $1, 2, \cdots, m$. The prior probability of the object being in box i is π_i, $\Sigma \pi_i = 1$. A search in box i for Δs time units costs $c_i \Delta s$ and results in a probability $\alpha_i \Delta s + o(\Delta s)$ of finding the object if it is there. The parameters $c_i > 0, \alpha_i > 0$ and $\pi_i > 0$, $i = 1, 2, \cdots, m$ are all assumed fixed and known. The objective is to search the boxes until the object is found with a minimal expected total cost.

The basic model for a moving target assumes in addition that the object occasionally moves from one box to another, typically in accordance to some specified continuous-time Markov process with m states. In this paper we assume that the movement of the target is independent of our search policy. Other models studied allow for different movements such as 'intelligent' targets which try to avoid being detected (see for example Dobbie (1975)).

As basic references to the theory of optimal search the reader may consult the books by Stone (1975) and Gal (1980). Numerous papers have been devoted to the subject. For some of the more recent ones the reader is referred to Assaf and Zamir (1985), (1987),

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* Postal address for both authors: Department of Statistics, The Hebrew University, 91905 Jerusalem, Israel.
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Brown (1980), Dobbie (1974), Eagle (1984), Pollock (1970), Schweitzer (1971), Sharlin (1987), (1990), Stone and Richardson (1974), Stone (1977), (1979), Stone and Kadane (1981), Stromquist and Stone (1981), Washburn (1980). A paper of particular importance in understanding our results is Weber (1986).

In this paper we deal with the case $m = 2$, although some of the results of Section 2 can easily be extended to any m. The notation in this case is simplified by using π and $1-\pi$ rather than π_1 and π_2. The Markov process governing the movement of the object is parameterized by the two rates $\lambda_1, \lambda_2 > 0$. Thus whenever the object is in box 1(2), it stays there for an exponential length of time with mean $\lambda_1^{-1}(\lambda_2^{-1})$ and then moves to box 2(1).

Models studied in the literature so far assume a given 'search effort', scaled to equal unity, which may be used to search in either of the boxes at any time. To fix ideas, think of a person or a radar scanning device performing the search. More generally, the search effort may be split between the boxes so that $\nu_1 + \nu_2 = 1$. Applying this allocation for Δs time units is naturally assumed to cost $(c_1\nu_1 + c_2\nu_2)\Delta s$ and to result in a probability of $[\alpha_1\nu_1\pi + \alpha_2\nu_2(1-\pi)]\Delta s + o(\Delta s)$ of finding the object. Let T be the actual (random) time at which the object is found. A slight generalization of the model studied by Weber (1986) suggests the objective of minimizing the total expected cost given by

$$(1) \qquad E\int_0^T [c_0 + c_1\nu_1(s) + c_2\nu_2(s)]ds$$

where $c_0 > 0$ is the cost for each unit of real time gone by before the object is found. The two other terms in (1) obviously represent the costs of the search times in the two boxes.

The distinction between 'real time' and 'search time' is of major importance in this paper. In practice a large c_0 (e.g. the cost of delay in detecting an enemy submarine) will cause us to try and find the object very fast, even at the price of spending a lot of search effort.

It should be noted that in the basic model described at this stage, real time and search time are equal. To further clarify this denote $c_1^* = c_0 + c_1$, $c_2^* = c_0 + c_2$. Since $\nu_1(s) + \nu_2(s) \equiv 1$, (1) may equivalently be written as

$$E\int_0^T [c_1^*\nu_1(s) + c_2^*\nu_2(s)]ds$$

thereby eliminating the c_0 term.

Weber (1986) proves that the optimal search policy for this problem is of the following two-region form: there exists a point $0 \leqq \pi^* \leqq 1$ such that $\nu_1(s) = 1$ for $\pi > \pi^*$ and $\nu_2(s) = 1$ for $\pi < \pi^*$. It should be noted at this stage that the value of π changes during the search process to its current posterior value in accordance to formulas given in Section 2.

In this paper the following 'dynamic' generalization of the problem is studied. Suppose we may apply *any* search effort in each of the two boxes. Naturally, doubling the effort will double the search cost on one hand, but will also double (approximately) the probability of finding the object during a short search period. From a practical point of view this may mean searching with a group of people rather than just one person, or devoting several scanning devices or more memory space. The dynamic search model thus addresses not only the question of where to search but also what total effort of search should be exerted at any given time.

Mathematically, this merely changes the problem to that of minimizing (1) subject to $\nu_1(s) \geqq 0$, $\nu_2(s) \geqq 0$ and no other constraints. Intuitively, since the object is moving between the two boxes, it is apparent that in the dynamic version there may be times in which we will not search at all. For example, suppose $c_1 \gg c_2$, $\lambda_1 \gg \lambda_2$ and $\alpha_1 \ll \alpha_2$. If the current value of π is high it may be best to wait a bit until we are fairly sure it has moved to box 2 and search there where the search is cheaper and the chances of finding it are higher. As will be seen, this may indeed be the case.

The optimal search policy for the dynamic problem turns out to be a $0-\infty$ one: at any given time, either apply all the search effort you can in a given box or do not search at all. In more intuitive terms, this tells us to exert the effort only when it is worth while to do so and, if it is, to put in all the effort we can. A heuristic derivation of this property is given in Section 2. Several basic properties and derivations resulting from this policy are given in Section 3. For references to the dynamic programming techniques and results used the reader may consult basic books on the subject such as Fleming and Rishel (1975), Ross (1983), Whittle (1982), and Varaiya (1972).

Once the $0-\infty$ nature of the optimal policy is deduced it is reasonably conjectured that it has a three-region form: there exist $0 \leqq \pi^* \leqq \pi^{**} \leqq 1$ such that $(\nu_1(s), \nu_2(s))$ equals $(\infty, 0)$ for $\pi > \pi^{**}$, equals $(0, 0)$ for $\pi^* \leqq \pi \leqq \pi^{**}$, and equals $(0, \infty)$ for $\pi < \pi^*$. This conjecture however is false and it turns out that in the most general case the optimal policy is a five-region one. Depending on the parameters, the optimal policy may also consist of two, three or four regions. The various cases are presented in Section 4, which also includes a rigorous proof of optimality for one of the five-region cases.

Some numerical aspects and results are discussed in Section 5. Examples are given showing that all possibilities — two, three, four and five regions — exist. Numerical comparisons between the solution to the dynamic version and the model studied by Weber (with $\nu_1 + \nu_2 = 1$) show that the dynamic policy can be significantly more efficient. The saving in cost may be only a few percent in some cases, but in others the cost may be drastically reduced, and the cost becomes only a fraction of the orginal. In the most extreme cases of the parameters this fraction may in fact tend to zero, indicating that the relative efficiency of the dynamic method tends to infinity.

Some concluding comments are discussed in Section 6, and several technical details are deferred to the appendix.

2. Heuristic arguments and preliminary results

As indicated in Section 1, the problem studied in this paper is the dynamic version of search for a moving target, namely minimizing (1) subject to $\nu_1(s)$, $\nu_2(s) \geqq 0$. In the first part of this section we present the law of motion for the state variable. Using heuristic arguments, it is shown that the optimal values for the control variables are 0 or ∞. Motivated by these heuristic results, we present a transformation which eliminates the difficulty of unbounded controls, thus allowing a rigorous treatment of the problem in Sections 3 and 4.

Let $\pi(s)$, $s \geqq 0$, be the posterior probability that the object is in box 1, given our unsuccessful search history up to time s. It is assumed that $\pi(0)$ is fixed and known. The current value of $\pi(s)$ may be considered as the state at time s. Given $\pi(s) = \pi$ and our search effort in the two boxes (v_1, v_2) for the time period $(s, s + \Delta s)$ the value of $\pi(s + \Delta s)$ is computed using elementary arguments as

$$\begin{aligned}\pi(s+\Delta s) = \{&\pi[1-\alpha_1 v_1 \Delta s + o(\Delta s)][1-\lambda_1\Delta s + o(\Delta s)]\\ &+(1-\pi)[1-\alpha_2 v_2\Delta s + o(\Delta s)][\lambda_2 \Delta s + o(\Delta s)]\}\\ &\times\{1-\{[\alpha_1 v_1\pi + \alpha_2 v_2(1-\pi)]\Delta s + o(\Delta s)\}\}^{-1}.\end{aligned} \tag{2}$$

Taking the limit as $\Delta s \to 0$ we obtain

$$\dot{\pi}(s) = (\alpha_2 v_2 - \alpha_1 v_1)\pi(1-\pi) + \lambda_2(1-\pi) - \lambda_1 \pi. \tag{3}$$

Equation (3) is composed of two parts. The first is a result of our search effort. For the second, use the notation

$$\lambda(\pi) \equiv \lambda_2(1-\pi) - \lambda_1\pi = \lambda_2 - (\lambda_1+\lambda_2)\pi, \tag{4}$$

which represents the 'natural drift' due to the movement of the target.

Let $H(\pi)$ be the optimal value function for the problem with $\pi(0) = \pi$, that is

$$H(\pi) = \inf E\left\{\int_0^T [c_0 + c_1 v_1(s) + c_2 v_2(s)]ds \mid \pi(0) = \pi\right\} \tag{5}$$

where the infimum is taken over all possible search functions $v_1(s)$, $v_2(s)$. Standard arguments of dynamic programming yield the following optimality equation for H:

$$\begin{aligned}H(\pi) = \inf_{v_1, v_2 \geqq 0} [&(c_0 + c_1 v_1 + c_2 v_2)\Delta s\\ &+\{1-[\alpha_1 v_1 \pi + \alpha_2 v_2(1-\pi)]\Delta s + o(\Delta s)\}H(\pi + \Delta\pi)].\end{aligned} \tag{6}$$

Now write $H(\pi + \Delta\pi)$ as

$$H(\pi + \Delta\pi) = H(\pi) + H'(\pi)\dot{\pi}\Delta s + o(\Delta s) \tag{7}$$

to obtain, as $\Delta s \to 0$,

$$\begin{aligned}0 = \inf_{v_1, v_2 \geqq 0} \{&v_1[c_1 - \alpha_1\pi H(\pi) - \alpha_1\pi(1-\pi)H'(\pi)]\\ &+ v_2[c_2 - \alpha_2(1-\pi)H(\pi) + \alpha_2\pi(1-\pi)H'(\pi)] + c_0 + \lambda(\pi)H'(\pi)\}.\end{aligned} \tag{8}$$

Since Equation (8) is linear in both v_1 and v_2 it follows that the optimal values of v_1 and v_2 at π are both either 0 or ∞ depending on the signs of the appropriate quantities.

Evidently, values of $v_i = \infty$ need to be correctly interpreted, both mathematically and 'physically'. A natural interpretation is to consider the bounded version $0 \leqq v_i \leqq M$ and take the limit as $M \to \infty$. It turns out that it is mathematically more convenient to apply a transformation of the control variables into the compact set $[0, 1] \times [0, 1]$. The details are as follows:

for any $0 \leqq v_i < \infty$, $\Delta s > 0$, let

$$T(v_1, v_2, \Delta s) = (u_1, u_2, \Delta t)$$

with u_1, u_2 and Δt given by

$$u_i = \frac{v_i}{v_1 + v_2 + 1}, \qquad i = 1, 2,$$

$$\Delta t = (v_1 + v_2 + 1)\Delta s. \tag{9}$$

Note that $0 \leqq u_1 + u_2 < 1$, $\Delta t > 0$, and also

$$u_i \Delta t = v_i \Delta s, \qquad i = 1, 2,$$

$$u_1 + u_2 + \frac{\Delta s}{\Delta t} = 1.$$

Rewrite the problem in terms of the new variables u_1, u_2, and Δt, namely: the cost during the time interval Δt,

$$(c_0 + c_1 v_1 + c_2 v_2)\Delta s = [c_0(1 - u_1 - u_2) + c_1 u_1 + c_2 u_2]\Delta t; \tag{10}$$

the probability of finding the object during the time interval Δt

$$[\alpha_1 v_1 \pi + \alpha_2 v_2(1 - \pi)]\Delta s + o(\Delta s) = [\alpha_1 u_1 \pi + \alpha_2 u_2(1 - \pi)]\Delta t + o(\Delta t); \tag{11}$$

and the law of motion for π is given by

$$\dot{\pi}(u_1, u_2) = (\alpha_2 u_2 - \alpha_1 u_1)\pi(1 - \pi) + \lambda(\pi)(1 - u_1 - u_2). \tag{12}$$

The transformed problem readily extends to include the values $u_i = 1$ (corresponding to $v_i = \infty$ in the original presentation). We thus consider the problem of finding control functions $u_1(t), u_2(t) \geqq 0$, $0 \leqq u_1(t) + u_2(t) \leqq 1$, which minimize the total expected cost until the object is found. The derivation of the value function for this problem, denoted by F, is carried out along the same lines used for the function H to yield the optimality equation

$$\begin{aligned} 0 = \inf_{\substack{0 \leqq u_1 + u_2 \leqq 1 \\ 0 \leqq u_1, u_2}} \{ & u_1[c_1 - c_0 - \alpha_1 \pi F(\pi) - \alpha_1 \pi(1 - \pi)F'(\pi) - \lambda(\pi)F'(\pi)] \\ & + u_2[c_2 - c_0 - \alpha_2(1 - \pi)F(\pi) + \alpha_2 \pi(1 - \pi)F'(\pi) - \lambda(\pi)F'(\pi)] \\ & + c_0 + \lambda(\pi)F'(\pi)\}. \end{aligned} \tag{13}$$

Remark 1. In the original representation the time Δs reflects 'real time'. The transformed problem is formulated in terms of 'search time' Δt. As is evident from (9), this search time equals the sum of the search efforts in both boxes, plus the real time. When no search is performed (in either of the boxes), then real time and search time are equal. During periods in which an infinite search effort is applied (i.e. $v_1 + v_2 \to \infty$ or $u_1 + u_2 \to 1$), search time is accumulated while real time is negligible.

Remark 2. A different interpretation of the transformed problem is as follows: think of Δt as the real time of the transformed problem. Denote the cost of an idle unit of search effort per unit time by c_0. Applying given values of u_1 and u_2 for Δt time units thus results in a total cost of

$$[c_0(1 - u_1 - u_2) + c_1 u_1 + c_2 u_2]\Delta t.$$

The rates for the target in this case are $\lambda_i(1 - u_1 - u_2)$. Note that in this interpretation the rates depend on the search effort. The dependence is such that intensive search lowers the rates, while no search results in the original rates. It is straightforward to verify that the optimal value function indeed satisfies (13).

Remark 3. The solution of the transformed problem is derived in Sections 3 and 4. Some care is needed in tracing the solution back to the original problem: a value of $u_i = 1$ should be understood as performing a control of v_i for Δs time units with $v_i \to \infty$, $\Delta s \to 0$ and $v_i \Delta s \to 0$.

The notation v_1, v_2, s, Δs for the original problem and u_1, u_2, t, Δt for the transformed one is used throughout the paper.

3. Analysis of the 0–1 policies

In this section we study the value function when 0–1 policies are applied. The results are summarized in three lemmata. For clarity of presentation the proofs are given after the statements of all three lemmata.

Let $\pi_0 = \lambda_2/(\lambda_1 + \lambda_2)$ be the point where the natural drift $\lambda(\pi)$ is zero. If no search is performed at π_0 then the state will remain π_0 for ever and the object will never be found, resulting in an infinite cost. Thus an interval where the control (0, 0) is applied will never contain π_0.

Lemma 1. Let $I = [a, b]$ be an interval in which a 0–1 search policy (u_1, u_2) is employed (i.e. $u_1(\pi) \equiv u_1$, $u_2(\pi) \equiv u_2$, both 0 or 1). The value function of the policy for $\pi \in (a, b)$ is of the form

$$L_1(\pi) = K_1(1 - \pi) - \frac{c_1}{\alpha_1}(1 - \pi)\ln\left(\frac{1 - \pi}{\pi}\right) + \frac{c_1}{\alpha_1} \tag{14}$$

if $(u_1, u_2) = (1, 0)$;

$$L_2(\pi) = K_2\pi - \frac{c_2}{\alpha_2}\pi \ln\left(\frac{\pi}{1 - \pi}\right) + \frac{c_2}{\alpha_2} \tag{15}$$

if $(u_1, u_2) = (0, 1)$;

$$L_3(\pi) = \begin{cases} L_3(a) + \dfrac{c_0}{\lambda_1 + \lambda_2} \ln\left(\dfrac{\lambda(\pi)}{\lambda(a)}\right), & \pi_0 < a \\ L_3(b) + \dfrac{c_0}{\lambda_1 + \lambda_2} \ln\left(\dfrac{\lambda(\pi)}{\lambda(b)}\right), & \pi_0 > b \end{cases} \tag{16}$$

if $(u_1, u_2) = (0, 0)$.

Also,

$$L_1'(\pi) = -K_1 + \frac{c_1}{\alpha_1 \pi} + \frac{c_1}{\alpha_1} \ln\left(\frac{1-\pi}{\pi}\right) \tag{17}$$

$$L_1''(\pi) = \frac{-c_1}{\alpha_1 \pi^2 (1-\pi)} \tag{18}$$

$$L_2'(\pi) = K_2 - \frac{c_2}{\alpha_2(1-\pi)} - \frac{c_2}{\alpha_2} \ln\left(\frac{\pi}{1-\pi}\right) \tag{19}$$

$$L_2''(\pi) = -\frac{c_2}{\alpha_2 \pi (1-\pi)^2} \tag{20}$$

$$L_3'(\pi) = \frac{-c_0}{\lambda(\pi)} \tag{21}$$

$$L_3''(\pi) = \frac{-c_0(\lambda_1 + \lambda_2)}{[\lambda(\pi)]^2}. \tag{22}$$

Consider now an arbitrary state $\pi_0 < x < 1$. There exists a unique search effort of the type $(0, u_2)$ for which the state will remain x for ever. The value of this control readily follows from (12) and is given by $|\lambda(x)|/(|\lambda(x)| + \alpha_2 x(1-x))$.

Lemma 2. Let

$$(u_1, u_2) = \left(0, \frac{|\lambda(x)|}{|\lambda(x)| + \alpha_2 x(1-x)}\right),$$

where $x > \pi_0$.

(a) Any search policy which uses the above control at state x has the value $L(x)$ at x given by

$$L(x) = \frac{c_2}{\alpha_2(1-x)} - \frac{c_0 x}{\lambda(x)}. \tag{23}$$

(b) If there exists an optimal policy which uses the above control at $\pi_2 > \pi_0$ then π_2 is the unique solution in $(\pi_0, 1]$ of the equation (in x)

$$\frac{x(1-x)^2}{(\lambda(x))^2} = \frac{c_2}{\alpha_2 c_0(\lambda_1 + \lambda_2)}. \tag{24}$$

This unique solution will henceforth be denoted by $\bar{\pi}_2$. The unique solution to (24) in $(0, \pi_0)$ will be denoted by $\underline{\pi}_2$. (See A1 in the appendix.)

The analogous form and result for $x < \pi_0$ are:

(a) Any search policy which uses the control

$$\left(\frac{\lambda(x)}{\lambda(x)+\alpha_1 x(1-x)}, 0\right)$$

at state x has the value $L(x)$ at x given by

$$L(x) = \frac{c_1}{\alpha_1 x} + \frac{c_0(1-x)}{\lambda(x)}. \tag{25}$$

(b) If there exists an optimal policy which uses the above control at $\pi_1 < \pi_0$ then π_1 is the unique solution in $[0, \pi_0)$ of the equation

$$\frac{x^2(1-x)}{(\lambda(x))^2} = \frac{c_1}{\alpha_1 c_0(\lambda_1+\lambda_2)}. \tag{26}$$

Denote this unique solution by $\bar{\pi}_1$ and the solution in $(\pi_0, 1)$ by $\underline{\pi}_1$ (see A2 in the appendix).

Note that, by (12), $\dot{\pi}(u_1, u_2) = 0$ for

$$(u_1, u_2) = \left(\frac{\alpha_2}{\alpha_1+\alpha_2}, \frac{\alpha_1}{\alpha_1+\alpha_2}\right).$$

For this case we have the following result.

Lemma 3. Let

$$(u_1, u_2) = \left(\frac{\alpha_2}{\alpha_1+\alpha_2}, \frac{\alpha_1}{\alpha_1+\alpha_2}\right).$$

(a) Any search policy which applies the above control at state x has the value $L(x)$ at x given by:

$$L(x) = \frac{c_1}{\alpha_1} + \frac{c_2}{\alpha_2}. \tag{27}$$

(The value does not depend on x.)

(b) If there exists an optimal policy which uses the above control at x then x is the solution of $(c_1/\alpha_1 x) = (c_2/\alpha_2(1-x))$ and is given by

$$x = \frac{c_1\alpha_2}{c_1\alpha_2 + c_2\alpha_1}. \tag{28}$$

This value will henceforth be denoted by π^*.

Remark 4. It should be noted that the controls suggested in Lemmata 2 and 3 are not of the 0–1 type, but they are equivalent to 0–1 controls. This equivalence will now be described for the case studied in Lemma 2: others may be described similarly.

When at x use $(u_1, u_2) = (0, 1)$ for Δt time units until the process reaches the state $x + \Delta x$ and then use $(u_1, u_2) = (0, 0)$ until the natural drift causes the process to reach x again. Repeating this procedure in the limiting $\Delta t \to 0$ sense will yield the same results as above. In terms of the process this amounts to noting that using a control which keeps us at x for ever is equivalent (in the limiting sense) to a policy which causes infinitesimal fluctuations around x.

Proof of lemmata. Let $L(\pi)$ be the value function for a given policy. Suppose that starting at state π the policy uses the controls (u_1, u_2) over the first time interval Δt. We have

$$\begin{aligned} L(\pi) &= (c_1u_1 + c_2u_2 + c_0(1 - u_1 - u_2))\Delta t \\ &\quad + (1 - (\alpha_1\pi u_1 + \alpha_2(1 - \pi)u_2)\Delta t + o(\Delta t))[L(\pi) + L'(\pi)\dot{\pi}\Delta t + o(\Delta t)]. \end{aligned} \tag{29}$$

Letting $\Delta t \to 0$ the value function satisfies the equation

$$\begin{aligned} 0 &= u_1[c_1 - \alpha_1\pi L(\pi) - \alpha_1\pi(1 - \pi)L'(\pi)] \\ &\quad + u_2[c_2 - \alpha_2(1 - \pi)L(\pi) + \alpha_2\pi(1 - \pi)L'(\pi)] \\ &\quad + (1 - u_1 - u_2)[c_0 + \lambda(\pi)L'(\pi)]. \end{aligned} \tag{30}$$

Think of (30) as

$$0 = u_1A_1 + u_2A_2 + (1 - u_1 - u_2)A_0 \tag{31}$$

where each A_i is a function of π, $L(\pi)$ and $L'(\pi)$.

Proof of Lemma 1. $(u_1, u_2) = (1, 0)$ implies, using (31), that $A_1 = 0$ and indeed L_1 specified in the statement of the lemma is the solution of the differential equation

$$c_1 - \alpha_1\pi L(\pi) - \alpha_1\pi(1 - \pi)L'(\pi) = 0. \tag{32}$$

In a similar way $(u_1, u_2) = (0, 1)$ and $(u_1, u_2) = (0, 0)$ imply $A_2 = 0$ and $A_3 = 0$ respectively resulting in L_2 and L_3 as the solutions to the appropriate differential equations, proving (b) and (c). The derivations of (17)–(22) are straightforward.

Proof of Lemma 2.

(a) When the control is as stated in the lemma then $\dot{\pi}(u_1, u_2) = 0$ so that the coefficient of $L'(\pi)$ vanishes and hence A_i of (31) are functions of π and $L(\pi)$ only. Using the above control results in the reduction of (31) to $0 = u_2A_2 + (1 - u_2)A_0$ or

$$|\lambda(x)|(c_2 - \alpha_2(1 - x)L(x)) + \alpha_2(1 - x)xc_0 = 0. \tag{33}$$

Recalling that $x > \pi_0$ and hence $\lambda(x) < 0$ the required result for $L(x)$ follows.

(b) The existence and uniqueness of a solution to (24) in $(\pi_0, 1]$ follows from Lemma A1 in the appendix. Let $\pi_2 > \pi_0$ be any point as in the lemma. The value of the policy at π_2 is $L(\pi_2)$ given in (23) and is assumed optimal. Hence $L(\pi_2)$ should be no greater than the value of any other policy at π_2. To prove (b), $L(\pi_2)$ is thus compared to the value function of two other policies at π_2. First consider the policy which at π_2 uses the control (0, 1) for Δt time units until the point $x = \pi_2 + \dot{\pi}(0, 1)\Delta t$ is reached and then at x the control

$$\left(0, \frac{|\lambda(x)|}{|\lambda(x)| + \alpha_2 x(1-x)}\right)$$

is applied forcing the process to stay at x. We have:

$$L(\pi_2) \leqq c_2\Delta t + (1 - \alpha_2(1 - \pi_2)\Delta t + o(\Delta t))L(\pi_2 + \dot{\pi}(0, 1)\Delta t).$$

Taking the Taylor expansion for $L(\pi_2 + \dot{\pi}(0, 1)\Delta t)$ and letting $\Delta t \to 0$ we obtain

$$0 \leqq c_2 - \alpha_2(1 - \pi_2)L(\pi_2) + \alpha_2\pi_2(1 - \pi_2)L'(\pi_2).$$

Substituting $L(\pi_2)$ from (23) and its derivative which is equal to

$$\frac{c_2}{\alpha_2(1 - \pi_2)^2} - \frac{c_0\lambda_2}{(\lambda(\pi_2))^2}$$

and rearranging, using (4) for $\lambda(\pi)$, we have

$$\frac{\pi_2(1 - \pi_2)^2}{(\lambda(\pi_2))^2} \leqq \frac{c_2}{\alpha_2 c_0(\lambda_1 + \lambda_2)} .$$

To prove the reverse inequality consider the policy which at π_2 applies the control (0, 0) for small Δt time units until the point $x = \pi_2 + \dot{\pi}(0, 0)\Delta t$ is reached, $x > \pi_0$ (note that $\dot{\pi}(0, 0) < 0$ for $\pi_2 > \pi_0$ but if Δt is small enough then $x > \pi_0$), and then stays at x by using

$$\left(0, \frac{|\lambda(x)|}{|\lambda(x)| + \alpha_2 x(1-x)}\right)$$

it follows that

$$L(\pi_2) \leqq c_0\Delta t + L(\pi_2 + \lambda(\pi)\Delta t).$$

Using the same arithmetic as before the required reversed inequality follows, completing the proof.

Proof of Lemma 3.

(a) Follows straightforwardly from substituting of $u_1 = \alpha_2/(\alpha_1 + \alpha_2)$, $u_2 = \alpha_1/(\alpha_1 + \alpha_2)$ in (30).

(b) Follows using similar arguments to those of 2(b).

4. Proof of optimality

As mentioned in the introduction, the optimal policy may consist of two, three four or five regions. The number of regions and the exact form as functions of the parameters are presented in Figures 1 and 2. Theorem 1 gives a detailed presentation and proof of the most complicated five-region case which is Case 3 in Figures 1 and 2. The remaining cases are similar and somewhat easier.

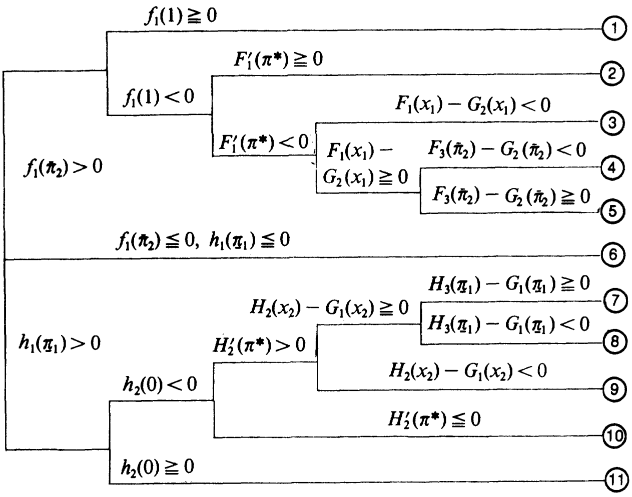

Figure 1

The five-region case arises when the parameters satisfy certain conditions. Altogether four conditions are needed, leading to two points which we denote by x_1 and y. Together with previously defined points (namely, $\bar{\pi}_2$ and π^*), x_1 and y determine the partition of the interval [0, 1] into the five regions.

The four conditions needed are not totally explicit, but are rather described in terms of the behavior of the functions L_1, L_2 and L_3 (i.e. the value functions in intervals where the controls used are (1, 0), (0, 1) and (0, 0) respectively). Recall that L_1, L_2 and L_3 are defined only up to some constants. These constants will now be determined, using boundary conditions which depend on the parameters, and the resulting function will compose the optimal value function (see (36)). We use F_i (as well as G_i and H_i when needed) to denote the function L_i with a suitable constant. The details are as follows.

Let $F_3(\pi)$ equal $L_3(\pi)$ with the boundary condition

$$F_3(\bar{\pi}_2) = \frac{c_2}{\alpha_2(1-\bar{\pi}_2)} - \frac{c_0\bar{\pi}_2}{\lambda(\bar{\pi}_2)} .$$

Thus for $\bar{\pi}_2 \leqq \pi \leqq 1$

$$F_3(\pi) = \frac{c_2}{\alpha_2(1-\bar{\pi}_2)} - \frac{c_0\bar{\pi}_2}{\lambda(\bar{\pi}_2)} + \frac{c_0}{\lambda_1+\lambda_2} \ln\left[\frac{\lambda(\pi)}{\lambda(\bar{\pi}_2)}\right]. \tag{34}$$

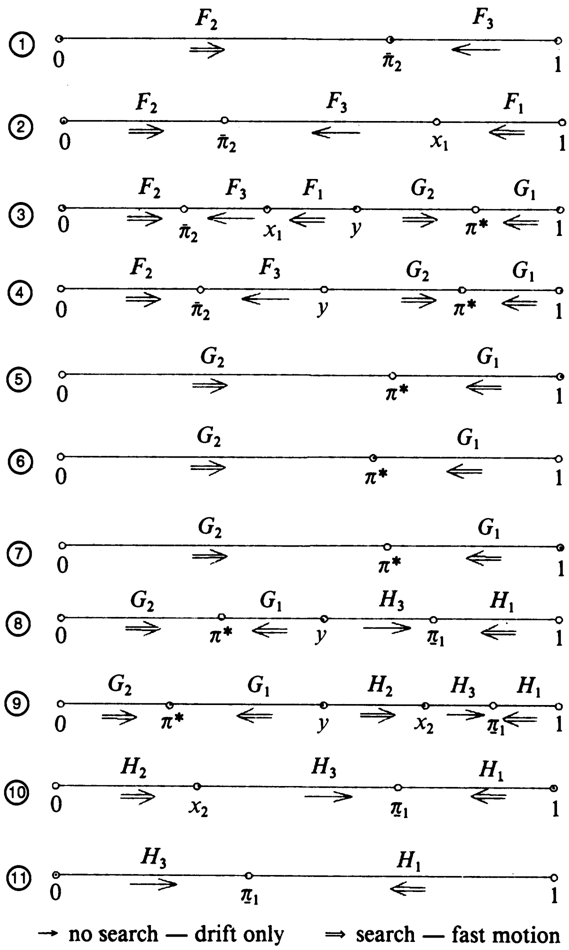

Figure 2

Define $f_1(\pi)$ for $\bar{\pi}_2 \leqq \pi \leqq 1$ as

$$f_1(\pi) = \frac{c_1}{\alpha_1 \pi} - F_3(\pi) - (1-\pi)F_3'(\pi), \tag{35}$$

and assume the first two conditions

$$f_1(1) < 0 \quad \text{and} \quad f_1(\bar{\pi}_2) > 0.$$

Let x_1 be the unique point in $(\bar{\pi}_2, 1)$ such that $f_1(x_1) = 0$ (existence and uniqueness are proved in Lemma A2 in the appendix). Define F_1 and F_2 as L_1 and L_2 respectively, with the boundary conditions

$$F_1'(x_1) = F_3'(x_1) = -\frac{c_0}{\lambda(x_1)}$$

$$F_2'(\bar{\pi}_2) = F_3'(\bar{\pi}_2) = -\frac{c_0}{\lambda(\bar{\pi}_2)}$$

and assume the third condition $F_1'(\pi^*) < 0$.

Now define G_1 and G_2 to equal L_1 and L_2 respectively, with the boundary conditions

$$G_1(\pi^*) = G_2(\pi^*) = \frac{c_1}{\alpha_1} + \frac{c_2}{\alpha_2}$$

and make the fourth (and last) assumption $F_1(x_1) < G_2(x_1)$. Let y be the unique point in (x_1, π^*) satisfying $F_1(y) = G_2(y)$ (see proof of Theorem 1 for existence and uniqueness of such a y).

Theorem 1. With the four assumptions and the preceding notation, the optimal value function is given by

$$F(\pi) = \begin{cases} F_2(\pi), & 0 \leqq \pi < \bar{\pi}_2 \\ F_3(\pi), & \bar{\pi}_2 \leqq \pi < x_1 \\ F_1(\pi), & x_1 \leqq \pi < y \\ G_2(\pi), & y \leqq \pi < \pi^* \\ G_1(\pi), & \pi^* \leqq \pi \leqq 1. \end{cases} \tag{36}$$

The value function in (36) is attained by the policy

$$(u_1, u_2)(\pi) = \begin{cases} (0, 1), & 0 \leqq \pi < \bar{\pi}_2 \\ (0, 0), & \bar{\pi}_2 \leqq \pi < x_1 \\ (1, 0), & x_1 \leqq \pi < y \\ (0, 1), & y \leqq \pi < \pi^* \\ (1, 0), & \pi^* \leqq \pi \leqq 1. \end{cases} \tag{37}$$

Remark 5. As evident from (37), the five-region case arises as a result of the insertion of the interval $[y, \pi^*)$ in which box 2 is to be searched, inside the 'large' interval $[x_1, 1]$ within which the 'natural' choice would be to search box 1 throughout. A numerical example in which this case occurs is when $c_1/\alpha_1 = 20$, $c_2/\alpha_2 = 3.7838$, $c_0 = 1.182425$, $\lambda_1 = 1$ and $\lambda_2 = 3$. For more details on the various cases, the reader is referred to the discussion on numerical results in Section 5.

Proof. We begin by showing that y exists and is unique. Direct computation shows that $G_1'(\pi^*) = G_2'(\pi^*) = 0$. Hence $F_1'(\pi^*) < G_1'(\pi^*)$ by the third condition. Since F_1 and G_1 are the same up to a constant it follows that $F_1'(\pi) < G_1'(\pi)$ for all π, so that $F_1(\pi) > G_1(\pi)$ for all π (see the exact form of $L_1(\pi)$), in particular $F_1(\pi^*) > G_1(\pi^*) = G_2(\pi^*)$. Since it is assumed that $F_1(x_1) < G_2(x_1)$ it follows that $x_1 < y < \pi^*$ exists. Uniqueness follows from the concavity of $F_1(\pi) - G_2(\pi)$ over the interval (x_1, π^*), which in turn may be shown by direct computation.

Applying the arguments of Section 3 it is easily verified that the value function in (36) is indeed attained using the policy in (37).

The proof of optimality is based on standard arguments in dynamic programming, namely showing that $F(\pi)$ satisfies the optimality equation for this case (see for example Varaiya (1972), Chapters 8 and 9). The optimality equation is derived in (13). Rearrange it to obtain

$$0 = \min_{\substack{0 \leqq u_1 + u_2 \leqq 1 \\ 0 \leqq u_1, u_2}} \{u_1[c_1 - \alpha_1 \pi F(\pi) - \alpha_1 \pi(1-\pi)F'(\pi)]$$

$$+ u_2[c_2 - \alpha_2(1-\pi)F(\pi) + \alpha_2\pi(1-\pi)F'(\pi)] \tag{38}$$

$$+ (1 - u_1 - u_2)[c_0 + \lambda(\pi)F'(\pi)]\}.$$

For $F(\pi)$ given in (36), one of the three terms in (38) always vanishes (since F solves the corresponding differential equation). The main technical difficulty is in proving that the remaining two terms are non-negative. Once this is done, it readily follows that the two corresponding controls (out of u_1, u_2 and $1 - u_1 - u_2$) attaining the minimum in (38) are zeros, matching the policy in (37). This argument thus simultaneously proves optimality of both the value function and the policy.

As an example take the interval $[0, \bar{\pi}_2)$. In this interval it thus remains to prove the pair of inequalities

(a) $c_1 - \alpha_1 \pi F_2(\pi) - \alpha_1 \pi(1-\pi)F_2'(\pi) \geqq 0$,

(b) $c_0 + \lambda(\pi)F_2'(\pi) \geqq 0$.

Similarly in the other regions it suffices to prove the following inequalities:

For $\bar{\pi}_2 \leqq \pi < x_1$

(c) $(c_1/\alpha_1\pi) - F_3(\pi) - (1-\pi)F_3'(\pi) \geqq 0$,

(d) $(c_2/\alpha_2(1-\pi)) - F_3(\pi) + \pi F_3'(\pi) \geqq 0$.

For $x_1 \leqq \pi < y$

(e) $(c_2/\alpha_2(1-\pi)) - F_1(\pi) + \pi F_1'(\pi) \geqq 0$,

(f) $c_0 + \lambda(\pi)F_1'(\pi) \geqq 0$.

For $y \leqq \pi < \pi^*$

(g) $(c_1/\alpha_1\pi) - G_2(\pi) - (1-\pi)G_2'(\pi) \geqq 0$,

(h) $c_0 + \lambda(\pi)G_2'(\pi) \geqq 0$.

And for $\pi^* \leqq \pi \leqq 1$

(i) $(c_2/\alpha_2(1-\pi)) - G_1(\pi) + \pi G_1'(\pi) \geqq 0$

(j) $c_0 + \lambda(\pi)G_1'(\pi) \geqq 0$.

The proofs of (a)–(j) are technical and follow from the four assumptions and the definitions of F_i, G_i, x_1 and y:

(a) First note that $F_2'(\bar{\pi}_2) = F_3'(\bar{\pi}_2)$ implies $F_2(\bar{\pi}_2) = F_3(\bar{\pi}_2)$. Hence $f_1(\bar{\pi}_2) > 0$ implies that (a) holds at $\pi = \bar{\pi}_2$. To complete the proof of (a) it thus suffices to show that $(c_1/\alpha_1\pi) - F_2(\pi) - (1-\pi)F_2'(\pi)$ is decreasing over $[0, \bar{\pi}_2)$. Differentiating, it readily follows that the function is decreasing over $[0, \pi^*)$ and it thus remains to show that in this case $\bar{\pi}_2 < \pi^*$. Now $f_1(\bar{\pi}_2) > 0$ implies, after some simple rearrangement,

$$\frac{c_1}{\alpha_1\bar{\pi}_2} - \frac{c_2}{\alpha_2(1-\bar{\pi}_2)} > -\frac{c_0}{\lambda(\bar{\pi}_2)} > 0$$

so that $\bar{\pi}_2 < \pi^*$ by the definition of π^*.

(b) Consider first the interval $[0, \pi_0)$. In this interval $\lambda(\pi) > 0$. Now $F_2'(\bar{\pi}_2) = -c_0/\lambda(\bar{\pi}_2) > 0$ and $F_2''(\pi) < 0$. Hence $F_2'(\pi) > 0$ for all $\pi < \bar{\pi}_2$ so that all terms in (b) are positive. In the interval $[\pi_0, \bar{\pi}_2)$, $\lambda(\pi) \leqq 0$. Since $c_0 + \lambda(\pi)F_3'(\pi) \equiv 0$ it suffices to show

$F_2'(\pi) \leqq F_3'(\pi)$ in this interval. The latter follows since $F_2'(\bar{\pi}_2) = F_3'(\bar{\pi}_2)$ and $F_2'(\pi) - F_3'(\pi)$ increases in $[\pi_0, \bar{\pi}_2]$ by Lemma A3 in the appendix.

(c) The function in (c) is, by definition, $f_1(\pi)$ which is non-negative over $[\bar{\pi}_2, x_1)$ by previous arguments and assumptions.

(d) By Lemma A4 in the appendix, the function in (d) attains its minimum in $(\pi_0, 1)$ at $\bar{\pi}_2$, and it thus suffices to show that

$$\frac{c_2}{\alpha_2(1-\bar{\pi}_2)} - F_3(\bar{\pi}_2) + \pi_2 F_3'(\bar{\pi}_2) \geqq 0.$$

However, $F_3'(\bar{\pi}_2) = F_2'(\bar{\pi}_2)$ implies $F_3(\bar{\pi}_2) = F_2(\bar{\pi}_2)$. The proof of (d) now follows since $c_2/\alpha_2(1-\pi) - F_2(\pi) + \pi F_2'(\pi)$ is identically zero.

(e) At $\pi = y$ we have

$$\frac{c_2}{\alpha_2(1-y)} - F_1(y) + yF_1'(y) \geqq \frac{c_2}{\alpha_2(1-y)} - G_2(y) + yG_2'(y) = 0,$$

where the left inequality holds since $F_1(y) = G_2(y)$ and $F_1'(y) \geqq G_2'(y)$ (since $F_1(\pi) > G_2(\pi)$ for all $y < \pi < \pi^*$, in particular, at the right-hand vicinity of y), and the right-hand equality follows from the definition of G_2.

For $x_1 \leqq \pi \leqq y$ it follows from straightforward differentiation that $(c_2/\alpha_2(1-\pi)) - F_1(\pi) + \pi F_1'(\pi)$ is decreasing (in fact it is decreasing over $(0, \pi^*)$).

(f) Begin with $c_0 + \lambda(\pi)F_3'(\pi) \equiv 0$. Since $\lambda(\pi) < 0$ in $[x_1, y]$, it suffices to show that $F_1'(\pi) \leqq F_3'(\pi)$. At x_1 we have equality, and $F_3' - F_1'$ increases over $(\bar{\pi}_1, 1)$ by Lemma A5 in the appendix. We now show $x_1 \geqq \bar{\pi}_1$. By Lemma A6 in the appendix $\bar{\pi}_1$ is the unique extremum point of f_1 in $(\pi_0, 1)$ which is a maximum so that $f_1(\bar{\pi}_1) > 0$ (since even $f_1(\bar{\pi}_2) > 0$). Since x_1 is the point where $f_1(x_1) = 0$ it thus follows that $x_1 \geqq \bar{\pi}_1$.

(g) At $\pi = \pi^*$ we have $G_2(\pi^*) = c_1/\alpha_1 + c_2/\alpha_2$ and $G_2'(\pi^*) = 0$. Thus

$$\frac{c_1}{\alpha_1 \pi^*} - G_2(\pi^*) - (1-\pi^*)G_2'(\pi^*) = 0$$

so that (g) holds as equality at π^*. The function $(c_1/\alpha_1\pi) - G_2(\pi) - (1-\pi)G_2'(\pi)$ is easily checked to be decreasing over (y, π^*), completing the proof.

(h) Since $\lambda(\pi) < 0$ and $c_0 + \lambda(\pi)F_3'(\pi) \equiv 0$ it suffices to show $G_2'(\pi) \leqq F_3'(\pi)$. Now $F_1'(y) \geqq G_2'(y)$ (see proof of (e)). Since $F_3'(x_1) = F_1'(x_1)$ and $F_3' - F_1'$ increases for $\pi > x_1$ $(> \bar{\pi}_1)$ by Lemma A5 in the appendix, it follows that $F_3'(y) \geqq F_1'(y) \geqq G_2'(y)$. The proof now follows since $F_3' - G_2'$ increases over (y, π^*) by Lemma A7.

(i) As in (g), (i) holds as equality at $\pi = \pi^*$ and $c_2/(\alpha_2(1-\pi)) - G_1(\pi) + \pi G_2'(\pi)$ increases over $(\pi^*, 1)$.

(j) The inequality holds since $G_1'(\pi^*) = 0$ and $G_1''(\pi^*) < 0$ so that $G_1'(\pi) < 0$ for $\pi > \pi^*$, while of course $\lambda(\pi) < 0$.

Remark 6. Some special care (which has been omitted in the proof) is needed at the points of connection $\bar{\pi}_2$, x_1, y and π^*, since both cases of $\Delta\pi > 0$ and $\Delta\pi < 0$ should be taken into account. This may be done by standard continuity arguments. For full details see Sharlin-Bilitzky (1990).

Remark 7. The following additional functions need to be defined to use Figures 1 and 2:

$$H_1(\pi) = B_1(1-\pi) - \frac{c_1}{\alpha_1}(1-\pi)\ln\left(\frac{1-\pi}{\pi}\right) + \frac{c_1}{\alpha_1},$$

$$B_1 = \frac{c_1}{\alpha_1 \underline{\pi}_1} + \frac{c_1}{\alpha_1}\ln\left(\frac{1-\underline{\pi}_1}{\underline{\pi}_1}\right) + \frac{c_0}{\lambda(\underline{\pi}_1)},$$

$$H_2(\pi) = B_2\pi - \frac{c_2}{\alpha_2}\pi\ln\left(\frac{\pi}{1-\pi}\right) + \frac{c_2}{\alpha_2},$$

$$B_2 = \frac{c_2}{\alpha_2(1-x_2)} + \frac{c_2}{\alpha_2}\ln\left(\frac{x_2}{1-x_2}\right) - \frac{c_0}{\lambda(x_2)}$$

(here x_2, to be defined below, is the analog of x_1).

$$H_3(\pi) = \frac{c_1}{\alpha_1 \underline{\pi}_1} + \frac{c_0(1-\underline{\pi}_1)}{\lambda(\underline{\pi}_1)} + \frac{c_0}{\lambda_1+\lambda_2}\ln\left(\frac{\lambda(\pi)}{\lambda(\underline{\pi}_1)}\right),$$

$$h_2(\pi) = \frac{c_2}{\alpha_2(1-\pi)} - H_3(\pi) - \frac{c_0\pi}{\lambda(\pi)}.$$

In the case where $h_2(0) < 0$ and $h_2(\underline{\pi}_1) > 0$, we define x_2 to be the unique point in $(0, \underline{\pi}_1)$ for which $h_2(x_2) = 0$.

5. Numerical aspects and results

A computer program which provides a full numerical solution for any values of the parameters has been written in PASCAL by Elazar Newman, and is available upon request. The program accepts the values of c_0, c_1, c_2, α_1, α_2, λ_1, λ_2 and an initial value $\pi(0)$ as input. The output consists of the number of regions, the values of the points of connection (e.g. x_1, y, π^*), the form of the optimal policy in each subinterval and the numerical value of the search starting at $\pi(0)$ $(F(\pi(0))$. Some, although not major, simplification is made by noticing that one needs only the values of c_1/α_1, c_2/α_2, $c_0/(\lambda_1+\lambda_2)$ and $\lambda_1/(\lambda_1+\lambda_2)$, thus reducing the number of parameters from seven to four.

We begin by showing that all cases (1)–(11) are indeed attained. Take $c_1/\alpha_1 = 20$, $c_2/\alpha_2 = 2.5$, $c_0 = 1.182425$, $\lambda_1 = 1$ and $\lambda_2 = 3$. The optimal policy in this case is a two-region one (case 1 in Figures 1 and 2 with $\bar{\pi}_2 = 0.8091$). This two region solution may well be anticipated intuitively since c_2/α_2 is much smaller than c_1/α_1.

Changing the value of one of the parameters will lead to the other cases. We demonstrate by increasing the value of c_2/α_2, indicating the various cases and some intuitive reasoning for each case as we go along.

Increase c_2/α_2 to 3.5 to obtain a three-region case (case 2 with $\bar{\pi}_2 = 0.8016$ and $x_1 = 0.8778$). Here a region where box 1 is searched is added. Note that increasing c_2/α_2 makes box 2 less favorable resulting in searching in box 1 as well in this case.

Next increase c_2/α_2 to 3.7838 to obtain the five-region case (case 3 with $\bar{\pi}_2 = 0.7999$, $x_1 = 0.8134$, $y = 0.8371$ and $\pi^* = 0.8409$). A further increase of c_2/α_2 to 3.785 results in a four-region case (case 4 with $\bar{\pi}_2 = 0.7999$, $y = 0.8113$, $\pi^* = 0.8409$. The values of $\bar{\pi}_2$ and π^* here are slightly, though not significantly different from those in the previous case.)

In case 3 an additional interval (G_2) in which an intensive search of box 2 is applied is inserted and in case 4 the F_1 interval vanishes. It might be surprising that increasing c_2/α_2 results in adding a region where box 2 is searched. A possible intuitive explanation is the following. The decisions through the search process are affected by both the hope for an immediate return (i.e., finding the object) and by future considerations in case of a failure. More specifically, in case of a failure we would like to control the trajectory of π wisely. Note that the state space is characterized by two types of stationary states: by stationary we mean states that once the searcher gets there he stays there until the object is found, as a result of the state and the optimal control applied. These states are π^* where both box 1 and box 2 are searched and $\bar{\pi}_2$ ($\underline{\pi}_1$) where a finite search effort is applied only in box 2(1).

Naturally increasing c_2/α_2 makes π^* a more favorable state to get 'stuck' in relatively to $\bar{\pi}_2$, and indeed the insertion of region G_2 in case 3 leads the searcher to π^* for some initial values of π. This phenomenon is more evident in case 4 where y decreases and F_1 vanishes. Nevertheless, in cases 3 and 4 there are still initial states from where the searcher will eventually get stuck in $\bar{\pi}_2$.

A further increase of c_2/α_2 to 3.7855 results in the two-region policy (G_1, G_2) (case 5) with a unique stationary point π^* ($\bar{\pi}_2$ plays no role any longer). Cases 6 and 7 are also two-region cases (as in case 5), and occur when c_2/α_2 is slightly increased. When $c_2/\alpha_2 = 10.465$ a four-region policy results with a new stopping region H_3 where the natural drift is towards $\underline{\pi}_1$ which now becomes an attractive stationary point (case 8). A five-region policy results for $c_2/\alpha_2 = 10.466$ widening the range of initial values of π with trajectory that leads to $\underline{\pi}_1$ ($\pi^* = 0.6565$, $y = 0.6821$, $x_2 = 0.6905$, $\underline{\pi}_1 = 0.7034$). $c_2/\alpha_2 = 15$ leads to case 10 and $c_2/\alpha_2 = 30$ to case 11 first making $\underline{\pi}_1$ a unique stationary point (case 10) and then degenerating region H_2.

The section concludes with some remarks regarding the efficiency of the dynamic search. Evidently, the solution of the dynamic procedure will always yield a smaller value than that of the basic constrained problem discussed by Weber (1986). Numerical trials indicate that the improvement may be minor in some cases while being most significant in others. It is not difficult to construct examples in which the dynamic procedure is highly superior: The optimal policy for the constrained problem is a two-region one which searches in box 2 for $0 \leqq \pi < \pi_{cr}$ for some π_{cr}. For proper choice of the parameters this policy will cause the process to remain forever at some fixed $\pi < \pi_{cr}$ and search only in box 2, resulting in a total expected cost of $(c_0 + c_2)/\alpha_2(1 - \pi)$ (see Weber (1986)).

Take this value of π and perform the (possibly suboptimal) dynamic policy of $(\nu_1, \nu_2) = (0, \infty)$ to the left of π and $(\nu_1, \nu_2) = (0, \infty)$ to the right of π. This will lead to a total expected cost of $(c_1/\alpha_1) + (c_2/\alpha_2)$ (see Lemma 3). Thus, if c_0 is large compared to the other relevant factors, the dynamic policy will yield a much lower expected cost. The relative cost fraction will in fact tend to zero as c_0/c_1 and c_0/c_2 tend to infinity.

6. Concluding remarks

1. In addition to the constraint $\nu_1 + \nu_2 = 1$, Weber (1986) also suggests considering the problem with an inequality constraint $\nu_1 + \nu_2 \leqq 1$. This problem is not solved by Weber who does however make the (very reasonable) conjecture that the optimal policy is a three-region one with $(\nu_1, \nu_2) = (0, 0)$ in the central interval.

The results of this paper show that the conjecture is false. To see this simply note that, by rescaling, the constraint may be written as $\nu_1 + \nu_2 \leqq M$. Let $M \to \infty$ (all other quantities kept fixed), to obtain the solution of the dynamic problem which, in general, is not a three-region one.

2. Some of the results of this paper can easily be generalized to more than two boxes. Most notably the $\nu = 0$ or $\nu = \infty$ types of control will remain optimal. This may be seen by writing the dynamic programming equations in a form analogous to (8). It is also possible to write and solve the differential equations in interior points of regions where such specific controls are used. Determining the number of regions and their boundaries for an optimal policy however seems to be an extremely difficult problem for $m > 2$, and at this stage no significant progress in this direction has been made.

Appendix

The following seven lemmata are needed for completing the proof of Theorem 1. Proofs are given for the first three. Some analogous lemmata are needed for the other cases described in Figures 1 and 2. For a full list of the lemmata needed as well as their detailed proofs the reader is referred to Sharlin-Bilitzky (1990).

Lemma A1. The function $\pi(1-\pi)^2/[\lambda(\pi)]^2$ increases in $[0, \pi_0)$ and decreases in $(\pi_0, 1]$.

Proof. Begin with $0 \leqq \pi < \pi_0$. Multiplying by $(\lambda_1 + \lambda_2)^2$ it suffices to prove the lemma for the function $\pi(1-\pi)^2/(\pi_0 - \pi)^2$. Differentiating and organizing terms, it now suffices to show that the function

$$(\pi_0 - \pi)(1 - 4\pi + 3\pi^2) + 2\pi(1-\pi)^2 \tag{A1}$$

is non-negative over $[0, \pi_0)$.

The quadratic term $1 - 4\pi + 3\pi^2$ has $\frac{1}{3}$ and 1 as its roots. Thus for $\pi \leqq \min(\pi_0, \frac{1}{3})$ the inequality is obvious. The inequality also holds at $\pi = \frac{1}{3}$ and $\pi = \pi_0$ and it thus remains to show that the derivative of (A1) does not vanish in $(\frac{1}{3}, \pi_0)$. Differentiate (A1) to obtain

$$-4\pi_0 + 6\pi_0\pi - 3\pi^2 + 1$$

whose discriminant is easily checked to be negative for $\frac{1}{3} < \pi_0 < 1$.

Analogously to (A1), it suffices to show that

$$(\pi - \pi_0)(1 - 4\pi + 3\pi^2) - 2\pi(1 - \pi)^2 \tag{A2}$$

is negative over $(\pi_0, 1]$. The function in (A2) is however no larger than $-(1 - \pi)^2(\pi_0 + \pi)$, which is of course negative.

Remark A1. The function in Lemma A1 obviously vanishes at $\pi = 0$ and $\pi = 1$ and tends to ∞ at π_0, so that $\underline{\pi}_2$ and $\bar{\pi}_2$ always exist and are unique.

Remark A2. A similar result holds for the function $\pi^2(1 - \pi)/[\lambda(\pi)]^2$.

Lemma A2. Assume $f_1(\bar{\pi}_2) > 0$ and $f_1(1) < 0$. Then there exists a unique x_1 in $(\bar{\pi}_2, 1)$ such that $f_1(x_1) = 0$.

Proof. Existence follows from the continuity of f_1 and the assumptions. Uniqueness will follow if we show that $f_1'(\pi)$ vanishes at most once in $(\bar{\pi}_2, 1)$ (at $\bar{\pi}_1$). Now

$$f_1'(\pi) = -\frac{c_1}{\alpha_1\pi^2} - (1 - \pi)F_3''(\pi) = -\frac{c_1}{\alpha_1\pi^2} + \frac{c_0(\lambda_1 + \lambda_2)(1 - \pi)}{[\lambda(\pi)]^2}$$

so that $f_1'(\pi) = 0$ if and only if

$$\frac{\pi^2(1 - \pi)}{[\lambda(\pi)]^2} = \frac{c_1}{\alpha_1 c_0(\lambda_1 + \lambda_2)}$$

which by Remark A2 has at most one solution in $(\bar{\pi}_2, 1)$, namely $\bar{\pi}_1$.

Lemma A3. The function $L_2'(\pi) - L_3'(\pi)$ increases over $(\pi_0, \bar{\pi}_2)$.

Proof. Differentiate to obtain

$$L_2''(\pi) - L_3''(\pi) = \frac{c_0(\lambda_1 + \lambda_2)}{[\lambda(\pi)]^2} - \frac{c_2}{\alpha_2(1 - \pi)^2\pi}$$

which is positive if and only if

$$\frac{(1 - \pi)^2\pi}{[\lambda(\pi)]^2} > \frac{c_2}{\alpha_2 c_0(\lambda_1 + \lambda_2)} .$$

The last inequality holds in $[\pi_0, \bar{\pi}_2)$ by Lemma A1.

Lemma A4. The function

$$\frac{c_2}{\alpha_2(1 - \pi)} - L_3(\pi) + \pi L_3'(\pi)$$

attains its minimum over the interval $[\pi_0, 1)$ at the point $\bar{\pi}_2$.

Lemma A5. The function $L_3'(\pi) - L_1'(\pi)$ is increasing over the interval $(\bar{\pi}_1, 1)$.

Lemma A6. The function $f_1(\pi)$ has a unique maximum over $(\pi_0, 1)$ at the point $\bar{\pi}_1$ (proof along the lines of Lemma A2).

Lemma A7. The function $L_3'(\pi) - L_2'(\pi)$ is increasing over $(\bar{\pi}_2, 1]$.

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